HVAC Load Forecasting in Office Buildings Using Machine Learning

Jingwen Zheng
State Grid Hubei Electric Power Research Institute
Wuhan, China
1092362588@qq.com

Zaixun Ling
State Grid Hubei Electric Power Research Institute
Wuhan, China
543762801@qq.com

Yiqun Kang
State Grid Hubei Electric Power Research Institute
Wuhan, China
376523814@qq.com

Li You
State Grid Hubei Electric Power Research Institute
Wuhan, China:
youli90909@163.com

Yukai Zhao
Hubei Huadian Innovation and New Energy Limited Company
Wuhan, China
414686289@qq.com

Ziwei Xiao
Huazhong University of Science and Technology
Wuhan, China
hustxiaoziwei@hust.edu.cn

Xiaofei Chen
Huazhong University of Science and Technology
Wuhan, China
chenxiaofei@hust.edu.cn

2021 6th International Conference on Power and Renewable Energy (ICPRE) | 978-1-6654-4961-8/21/$31.00 ©2021 IEEE | DOI: 10.1109/ICPRE52634.2021.9635310

***Abstract*—Accurate short-term load forecasting can be used to optimize the control and operation of the heating, ventilation and air-conditioning (HVAC) systems for energy conservation. Deep learning is becoming increasingly popular in load forecasting due to its excellent capacity to handle non-linear problems. In this study, a load forecasting model using long short-term memory (LSTM) neural network based on wavelet analysis is developed. This algorithm combines the ability of LSTM to capture time series characteristics and the ability of wavelet analysis to optimize data samples. The performance of the proposed method is compared with LSTM and back propagation neural network. The implementation of the proposed method in an office building shows that the model can realize the load forecast accurately with root mean squared errors of less than 124.7kW and the coefficient of variance of the root mean square errors (CVRMSE) of less than 24.4%. The CVRMSE of the proposed model is improved by 2.5% and 1.8% in summer and 2.3% and 0.3% in winter respectively, compared with back propagation neural network and traditional LSTM neural network.**

Keywords—LSTM, short-term, load prediction, back propagation, deep learning, HVAC

I. INTRODUCTION

The heating, ventilation and air-conditioning (HVAC) systems constitute over 50% of the total energy consumption of buildings [1]. Accurate HVAC load forecasting is a prerequisite for energy savings of HVAC systems, which could help reasonably arrange energy use and high-quality operation management for HVAC systems to ensure optimal operation and economic benefits [2].

The data-driven models applied in HVAC load forecasting are generally divided into statistical methods, artificial neural networks, support vector machines (SVM), deep learning and ensemble methods [3]. Reference [4] applied a multiple nonlinear regression model to forecast HVAC cooling load. The results showed the accuracy of the model is higher compared with traditional forecasting models. Reference [5] used degree-day as input features to forecast building HVAC energy consumption. This method reduced the computation time and was easily constructed with root mean square errors of less than 20% compared to previous models. Reference [6] combined deep reinforcement learning with autoencoder algorithm to forecast short-term HVAC system energy consumption. The results implied that the proposed method outperformed common supervised models such as back propagation (BP) Neural Network and SVM. Although the above algorithms have achieved accurate forecasting results, the time series characteristics are not fully considered.

Long short-term memory (LSTM) neural network belongs to recurrent neural network, which could deal with long-term information and is suitable for time series data prediction [7]. LSTM neural network is widely applied in load forecasting, such as day-ahead HVAC power consumption and plug-in loads. Reference [8] applied LSTM to forecast 24-h ahead of HVAC system load. The results showed the method provided accurate results with normalized root mean square error of 0.13. Reference [9] used LSTM to forecast different temporal granularities of electrical load. The results showed the proposed method outperformed traditional methods for all the forecasting temporal granularity.

Although LSTM neural network provides relatively accurate forecasting results, many optimization studies on it are conducted for further improvement on accuracy. The optimization of LSTM neural network generally includes the optimization of the LSTM models and the improvement on data preprocessing [10]. In terms of the LSTM model optimization, the traditional LSTM models can be combined with convolutional neural network (CNN) models for more accurate forecasting [11]. The CNN could extract complex features from inputs, which enhanced the mapping capability of the LSTM models. Reference [12] decomposed the original data into trend, seasonality and residual sublayers by a Naïve decomposition method. Independent LSTM models based on attention mechanism were constructed for every sublayer. The results

The research presented in this paper is supported by the National Natural Science Foundation of China (No. 51808238).

Authorized licensed use limited to: b-on: UNIVERSIDADE DA BEIRA INTERIOR. Downloaded on November 02,2026 at 16:25:23 UTC from IEEE Xplore. Restrictions apply.

showed the proposed models outperformed the simple LSTM model and attention-based LSTM model. In terms of the improvement in data preprocessing, the common methods include random forest (RF), partial least squares regression, wavelet analysis and principal components analysis (PCA), which could help choose suitable features or reconstruct feature vectors [13]. Reference [14] applied RF to select influential features and then forecast stock price based on LSTM neural network. The results showed the method could effectively select features and improve the performance of the forecasting model. Reference [15] applied PCA to extract key weather features for traffic flow forecasting. The results showed the extracted features enhanced the performance of LSTM models. Wavelet analysis is widely used to extract load features in various frequency domains and improve model performance, which has been applied to solar radiation forecasting and obtain good results [16-18].

On this basis, this paper develops a forecasting method based on LSTM neural network and wavelet analysis to forecast day-ahead HVAC load. LSTM neural network is applied to capture information on time series. Wavelet analysis is applied to decompose the time series data into low-frequency coefficients and high-frequency coefficients. A separate LSTM model is trained for every subsequence. Time index is considered in the method to provide temporal information such as day type. The performance of the proposed load forecasting model is compared with the BP AND LSTM models, which are widely used in HVAC load forecasting.

II. Model Structure

A. Long Short-term Memory Neural Network

Long Short-term Memory (LSTM) belongs to a variant of recurrent neural network, composed of gate units and memory cells [19]. Gate units are used to determine the information flows and update information in memory cells while memory cells store historical information. A typical structure of the LSTM unit is shown in Fig. 1. The calculation method of the LSTM unit is as shown in (1) - (5).

Forget gate

$$f_t = \sigma[\omega_f(x_t, h_{t-1}) + b_f] \tag{1}$$

Input gate

$$i_t = \sigma[\omega_i(x_t, h_{t-1}) + b_i] \tag{2}$$

$$\tilde{c}_t = \tanh[\omega_c(x_t, h_{t-1}) + b_c] \tag{3}$$

$$c_t = c_{t-1} f_t + \tilde{c}_t i_t \tag{4}$$

Output gate

$$h_t = \sigma[\omega_o(x_t, h_{t-1}) + b_o] \tanh c_t \tag{5}$$

where f_t represents the probability of information being forgotten at the last moment, x_t represents the input at the current time t, h_{t-1} represents the output at the last moment t-1, i_t represents the probability of the input information to save, $\tilde{c}_t$ represents the obtained information from the input at the current time, c_t represents the updated state value stored in the cell, c_{t-1} represents the last state value stored in the cell, and h_t represent the output of the neuron at the current moment. σ and tanh are the activation functions while ω and b are the weight and the bias of every neuron.

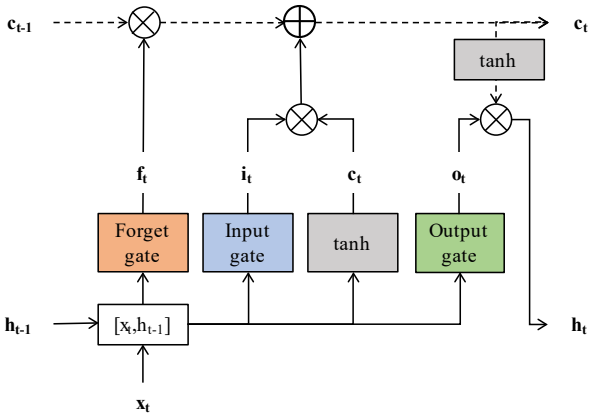


Fig. 1. A typical structure of the LSTM unit.

The LSTM neural network model in this study includes two LSTM layers: Dropout layer and Dense layer. The specific structure of the LSTM neural network model is shown in Fig. 2.

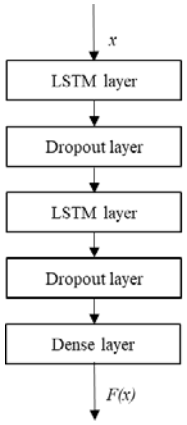


Fig. 2. Structure of the LSTM neural network model.

B. Wavelet Analysis

Wavelet analysis is a signal preprocessing approach, widely used in the field of load forecasting [20-22]. It can be used to reduce the influence of noisy data and extract features from hyperspectral data [23]. The original data are transformed into different frequent subseries and the extracted information better reflects the load characteristic [24]. Wavelet analysis transforms the load data into low- and high-frequency components based on wavelet bases. The low-frequency component provides an estimation of the general trend of the original signal while the high-frequency component represents detailed information [22]. The original series is decomposed as shown in (6):

$$y(t) = \sum_k S_{j_n,k} \varphi_{j_n,k}(t) + \sum_{j>j_n} \sum_k d_{j,k} \phi_{j,k}(t) \tag{6}$$

where $\varphi_{j_n,k}(t)$ and $\emptyset_{j,k}(t)$ represent the bases of scaling functions and wavelets, and $S_{j_n,k}$ and $d_{j,k}$ represents low-frequency and high-frequency coefficients.

Authorized licensed use limited to: b-on: UNIVERSIDADE DA BEIRA INTERIOR. Downloaded on November 02,2026 at 16:25:23 UTC from IEEE Xplore. Restrictions apply.

C. Process of Model Construction

The detailed flowchart of model construction is displayed in Fig. 3.

Step 1: Choose suitable bases of scaling functions and wavelets and decompose the original series into low-frequency and high-frequency components.

Step 2: Use LSTM models to forecast respectively low-frequency and high-frequency components based on historical low- and high-frequency data, temperature, humidity and day type (ie, one of the days from Monday to Sunday).

Step 3: Obtain the next-day forecasting results of the HVAC load by summing the next-day forecasting results for low-frequency and high-frequency components.

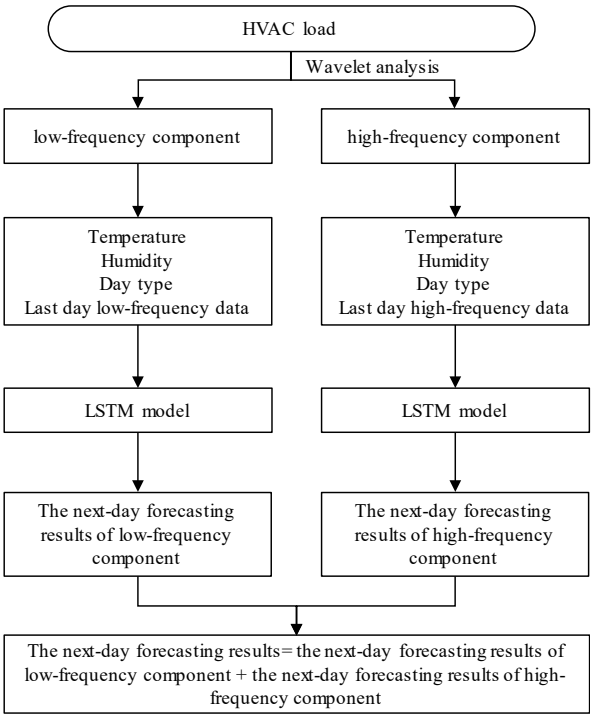


Fig. 3. Flowchart of model construction.

III. Experiments and Results

A. Data Preprocessing and Evaluation Indicators

The study implements the proposed approach in the real power data of an office building in Shanghai [25]. The dataset includes the load data for three successive years, with a sampling rate of 1h. The office building has twenty-nine floors above the ground, which provides a total indoor area of approximately 31,800 m^2. The annual hourly HVAC load, collected by smart meters, is introduced in Fig.4. The HVAC load is lower in the transition season while it is higher in the winter and summer. The load in summer gradually decreases before rising while the load in winter fluctuates greatly with peaks.

Considering the influence of schedule on occupant behavior and energy consumption, the day type is considered as the inputs. The day type is regarded as a categorical variable and 7 binary values represent the day of the week. In the study, the data from June to September for three successive years are selected as summer samples while the data from December to March are selected as winter samples. 80% of samples are selected as the train set and the remaining are selected as the test set. The data are normalized between 0 and 1 owing to the sensitivity of neural networks to data [26]. The method of data normalization is shown in (7):

$$X_i' = \frac{X_i - X_{min}}{X_{max} - X_{min}} \tag{7}$$

where X_i represents the original variables and $X_i^{'}$ is the normalized variables. X_{min} and X_{max} are the relevant minimum and maximum values of X_i, respectively.

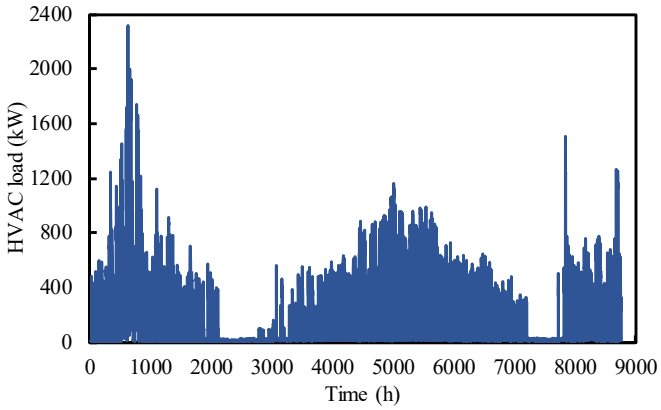


Fig. 4. Annual hourly HVAC load of the office building.

To evaluate the performance of the proposed model, root mean squared error (RMSE) and coefficient of variance of the root mean square error (CVRMSE) are used evaluation indexes as shown in (8) and (9):

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(\hat{y}_i - y_i)^2} \tag{8}$$

$$CVRMSE = \frac{\sqrt{\frac{1}{N}\sum_{i=1}^{N}(\hat{y}_i - y_i)^2}}{\bar{y}} \tag{9}$$

where $\hat{y}_i$ and y_i are the forecasting and actual value of the *i*-th sampling point, $\bar{y}$ is the average of the actual value and N is the number of samples.

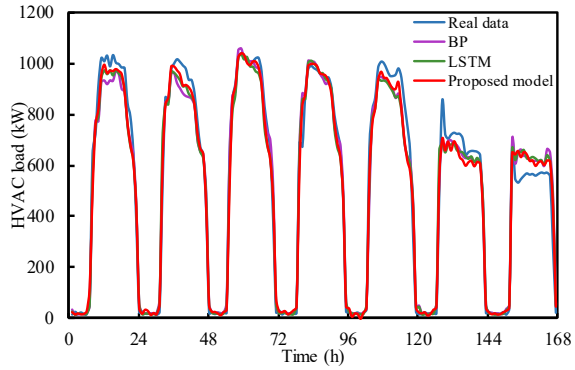
B. Analysis of Prediction Result

The study compares BP neural network, traditional LSTM neural network and LSTM neural network based on wavelet analysis to verify the effectiveness of the proposed model. The HVAC loads in summer and winter are forecasted in individual load forecasting models due to different load characteristics in winter and summer. Historical data, temperature, humidity and day type are considered as inputs in the study. In the study, both BP neural network and LSTM neural network have two hidden layers. The number of neurons in the two layers is both set to 200 as the optimal results of the number of neurons. The wavelet coefficients are extracted based on db2 with level 1.

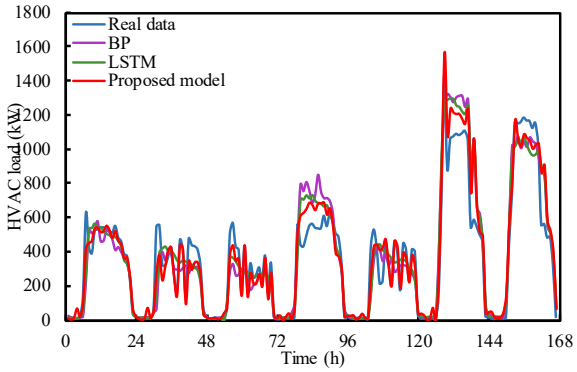
The forecasting results in a typical week are shown in Fig. 5. It can be seen the forecasting results of the proposed model

Authorized licensed use limited to: b-on: UNIVERSIDADE DA BEIRA INTERIOR. Downloaded on November 02,2026 at 16:25:23 UTC from IEEE Xplore. Restrictions apply.

match the real data better than the other models, indicating the proposed approach could improve the forecasting accuracy. When the HVAC load fluctuates severely, the errors will be large. However, there is a more similar trend between the real data and the proposed models than other models because the proposed model is more sensitive to change in time series data than other models. The results indicate the proposed models have better mapping approximation ability for data with severe fluctuation.



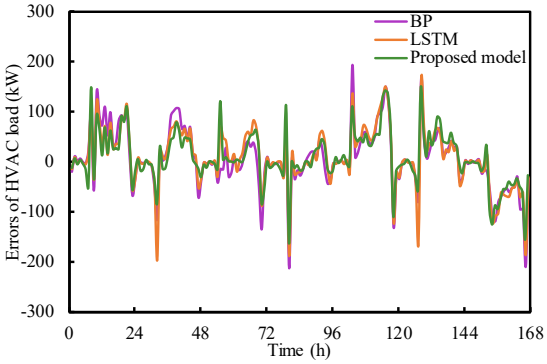
(a) Forecasting results in a typical summer week



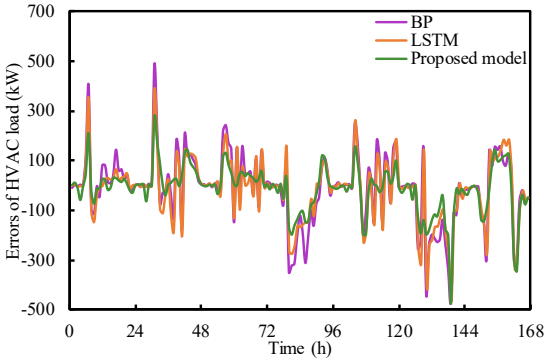
(b) Forecasting results in a typical winter week

Fig. 5. Comparison of different models in a typical week.

The errors of the forecasting results for the three models in a typical week are shown in Fig. 6. It can be seen that the proposed forecasting model outperforms the other models in most cases, especially at peaks or large fluctuations. The maximum errors of three forecasting models usually appear in the eighth hour for summer and in the seventh hour for winter. For the BP, LSTM and LSTM based on wavelet analysis forecasting models, the errors in summer range between -211.9kW and 194.6kW, -196.3kW and 166.8kW, and -161.3kW and 150.4kW, respectively. The errors in winter range between -471.6kW and 488.8kW, -465.4kW and 388.2kW, and -474.9kW and 278.1kW, respectively. The errors in winter are larger in winter than in summer. The possible reason is that load in winter fluctuates severely and difficult to forecast. During the process of rises and falls, the forecasting results are more accurate. The large errors are mainly produced when the load fluctuates severely or peaks. The proposed forecasting model fits the peaks and data with severe fluctuation better than the other two forecasting models.



(a) Errors in a typical summer week



(b) Errors in a typical winter week

Fig. 6. Comparison of errors of different models in a typical week.

Table I demonstrates the evaluation indexes of BP neural network, traditional LSTM neural network and LSTM neural network based on wavelet analysis for the testing set. The results show the LSTM neural network based on wavelet analysis outperforms the other models. It can be seen that the proposed model has the lowest RMSE of 61.6kW in summer and that of 124.7kW in winter. Compared to other models, the proposed model has the most accurate results, whose CVRMSEs reduce by 2.5% and 1.8% in summer and 2.3% and 0.3% in winter. The proposed method is slightly better in summer than that in winter.

TABLE I. COMPARISON OF EVALUATION INDEX OF DIFFERENT MODELS IN THE TESTING SET

Model	Summer		Winter	
	RMSE (kW)	*CVRMSE (%)*	*RMSE (kW)*	*CVRMSE (%)*
BP	76.1	17.7%	133.2	26.7%
LSTM	73.3	17.1%	128.1	24.7%
Proposed model	61.6	15.2%	124.7	24.4%

IV. CONCLUSION

Accurate forecasting of HVAC load contributes to optimal operation and energy savings of HVAC systems. In the paper, LSTM neural network based on wavelet analysis is developed to forecast day-ahead HVAC load. The method applies LSTM

Authorized licensed use limited to: b-on: UNIVERSIDADE DA BEIRA INTERIOR. Downloaded on November 02,2026 at 16:25:23 UTC from IEEE Xplore. Restrictions apply.

neural network to capture information on low- and high-frequency time series components based on wavelet analysis. The results show that the method can obtain more accurate forecasting results than BP neural network and traditional LSTM neural network models.

ACKNOWLEDGMENT

The research presented in this paper is supported by the National Natural Science Foundation of China (No. 51808238).

REFERENCES

[1] Pérez-Lombard L, Ortiz J, Pout C, "A review on buildings energy consumption information", Energy and Buildings, vol. 40, pp. 394-8, 2008.

[2] Fan C, Liao Y, Zhou G, Zhou X, Ding Y, "Improving cooling load prediction reliability for HVAC system using Monte-Carlo simulation to deal with uncertainties in input variables", Energy and Buildings, vol. 226, pp. 110372, August 2020.

[3] Zhao H-x, Magoulès F, "A review on the prediction of building energy consumption", Renewable and Sustainable Energy Reviews, vol. 16, pp. 3586-92, April 2012.

[4] Fan C, Ding Y, "Cooling load prediction and optimal operation of HVAC systems using a multiple nonlinear regression model", Energy and Buildings, vol. 197, pp. 7-17, May 2019.

[5] Sha H, Xu P, Hu C, Li Z, Chen Y, Chen Z, "A simplified HVAC energy prediction method based on degree-day", Sustainable Cities and Society, vol. 51, pp. 101698, July 2019.

[6] Liu T, Xu C, Guo Y, Chen H, "A novel deep reinforcement learning based methodology for short-term HVAC system energy consumption prediction", International Journal of Refrigeration, vol. 107, pp. 39-51, July 2019.

[7] Gao Y, Ruan Y, Fang C, Yin S, "Deep learning and transfer learning models of energy consumption forecasting for a building with poor information data", Energy and Buildings, vol. 223, pp. 110156, June 2020.

[8] Sendra-Arranz R, Gutiérrez A, "A long short-term memory artificial neural network to predict daily HVAC consumption in buildings", Energy and Buildings, vol. 216, pp. 109952, March 2020.

[9] Muzaffar S, Afshari A. "Short-Term Load Forecasts Using LSTM Networks." The 10th International Conference on Applied Energy, Hong Kong, vol. 158, pp. 2922-7, August 2018.

[10] Haiyan W, Xinhang L, Xiaonan L. "Short-term Load Forecasting of Power grid Based on Improved WOA Optimized LSTM." 2020 5th International Conference on Power and Renewable Energy (ICPRE), pp. 54-60, September 2020.

[11] Guo X, Zhao Q, Zheng D, Ning Y, Gao Y. "A short-term load forecasting model of multi-scale CNN-LSTM hybrid neural network considering the real-time electricity price." 7th International Conference on Power and Energy Systems Engineering, Fukuoka, Japan, vol. 6, pp. 1046-53, September 2020.

[12] Heidari A, Khovalyg D, "Short-term energy use prediction of solar-assisted water heating system: Application case of combined attention-based LSTM and time-series decomposition", Solar Energy, vol. 207, pp. 626-39, July 2020.

[13] Sun Y, Haghighat F, Fung BCM, "A review of the-state-of-the-art in data-driven approaches for building energy prediction", Energy and Buildings, vol. 221, pp. 110022, May 2020.

[14] Ma Y, Han R, Fu X. "Stock prediction based on random forest and LSTM neural network" 2019 19th International Conference on Control, Automation and Systems (ICCAS), pp. 126-30, October 2019.

[15] Mingzhou W, Shan T, Chao C, Jiandan Z. "Short-time traffic flow forecast with weather characteristics" 2020 International Conference on Computer Communication and Network Security (CCNS), pp. 142-5, 2020.

[16] Cao J, Lin X, "Application of the diagonal recurrent wavelet neural network to solar irradiation forecast assisted with fuzzy technique", Engineering Applications of Artificial Intelligence, vol. 21, pp. 1255-63, 2008.

[17] Capizzi G, Napoli C, Bonanno FJITNNLS, "Innovative Second-Generation Wavelets Construction With Recurrent Neural Networks for Solar Radiation Forecasting", EEE Transactions on Neural Networks and Learning Systems, vol. 23, pp. 1805-15, 2012.

[18] Zhang P, Takano H, Murata J. "Daily Solar Radiation Prediction based on Wavelet Analysis." SICE Annual Conference 2011, Waseda University, Tokyo, Japan, pp. 2598-603, 2011.

[19] Sheng F, Jia L. "Short-Term Load Forecasting Based on SARIMAX-LSTM." 2020 5th International Conference on Power and Renewable Energy (ICPRE), pp. 90-4, September 2020.

[20] Gong M, Wang J, Bai Y, Li B, Zhang L, "Heat load prediction of residential buildings based on discrete wavelet transform and tree-based ensemble learning", Journal of Building Engineering, vol. 32, pp. 101455, April 2020.

[21] Doucoure B, Agbossou K, Cardenas A, "Time series prediction using artificial wavelet neural network and multi-resolution analysis: Application to wind speed data", Renewable Energy, vol. 92, pp. 202-11, February 2016.

[22] Bashir ZA, El-Hawary ME, "Applying Wavelets to Short-Term Load Forecasting Using PSO-Based Neural Networks", IEEE Transactions on Power Systems, vol. 24, pp. 20-7, 2009.

[23] Zaroubi S, Goelman G, "Complex denoising of MR data via wavelet analysis: Application for functional MRI", Magnetic Resonance Imaging, vol. 18, pp. 59-68, 2000.

[24] Ghayekhloo M, Menhaj MB, Ghofrani M, "A hybrid short-term load forecasting with a new data preprocessing framework", Electric Power Systems Research, vol. 119, pp. 138-48, September 2015.

[25] Xiao T, Xu P, Sha H, Chen Z, Gu J. EnergyDetective2020_dataset 2021.

[26] Ahmad MW, Mourshed M, Rezgui Y, "Trees vs Neurons: Comparison between Random Forest and ANN for high-resolution prediction of building energy consumption", Energy & Buildings, vol. 147, pp. S0378778816313937, April 2017.

Authorized licensed use limited to: b-on: UNIVERSIDADE DA BEIRA INTERIOR. Downloaded on November 02,2026 at 16:25:23 UTC from IEEE Xplore. Restrictions apply.